

CONSTRAINT DISCOVERY FROM ABORTED ACTIONS: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

We study robot planning and safety. The motivating bet is: Use aborted or interrupted actions as evidence for hidden constraints.. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For robot planning and safety, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2020; 2023; 2013; 2022; 2021; 2018). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

3 MECHANISM

Given state s and action a , the model returns an observed next-state prediction plus a finite atlas $B(s, a)$ of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

Variant	Split	Success	Tail risk
observed_only_wam	nominal	0.542	0.458
observed_only_wam	hidden_mode_shift	0.205	0.795
augmentation_as_data_wam	nominal	0.570	0.430
augmentation_as_data_wam	hidden_mode_shift	0.233	0.767
scalar_uncertainty_wam	nominal	0.645	0.355
scalar_uncertainty_wam	hidden_mode_shift	0.383	0.617
proposed_branch_mechanism	nominal	0.782	0.218
proposed_branch_mechanism	hidden_mode_shift	0.573	0.427

5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

6 CONCLUSION

Constraint Discovery From Aborted Actions argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

REFERENCES

- Provably-correct stochastic motion planning with safety constraints, 2013. <https://doi.org/10.1109/icra.2013.6630692>.
- A constraint-based mission planning approach for reconfigurable multi-robot systems, 2018. <https://doi.org/10.4114/intartif.vol21iss62pp25-39>.
- Wasserstein distributionally robust motion planning and control with safety constraints using conditional value-at-risk, 2020. <https://doi.org/10.1109/icra40945.2020.9196857>.
- Learning-based end-to-end path planning for lunar rovers with safety constraints, 2021. <https://doi.org/10.3390/s21030796>.
- Laparoscopic automatic following motion planning of minimally invasive surgery robot based on safety constraints, 2022. <https://doi.org/10.1109/icma54519.2022.9856048>.
- Time optimal planning-based high-speed running motion generation for humanoid robots with safety constraints, 2023. <https://doi.org/10.23919/ccc58697.2023.10241009>.